

# HW 4

## STAT 5211

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a)

Using the **logistic function**,

```
# p(x) =  
exp(-6 + 0.05*40 + 1*3.5) / (1 + exp(-6 + 0.05*40 + 1*3.5))
```

```
## [1] 0.3775407
```

The probability of a student who studies for 40 hours and has an undergrad GPA of 3.5 getting an A in the class is approximately 0.3775 under the logistic model.

b)

Solving the **logit** function for  $X_1$ ,

```
# X1 =  
(log(0.5 / (1 - 0.5)) + 6 - 1*3.5) / 0.05
```

```
## [1] 50
```

The student in part (a) would need to study for 50 hours to have a 50% chance of getting an A in the class.

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a)

Consider the odds  $\frac{p(x)}{1-p(x)} = 0.37$ . Rearrange to solve for  $p(x)$ , that is,

```
# p(x) =  
.37 / 1.37
```

```
## [1] 0.270073
```

Approximately 27.01% of people with an odds of 0.37 of defaulting on their credit card payment will in fact default.

b)

Plugging in  $p(x) = 0.16$  into the odds,

```
# odds =  
0.16 / (1 - 0.16)
```

```
## [1] 0.1904762
```

The odds that an individual with a 16% chance of defaulting on her credit card payment *actually* defaulting are approximately 0.19.

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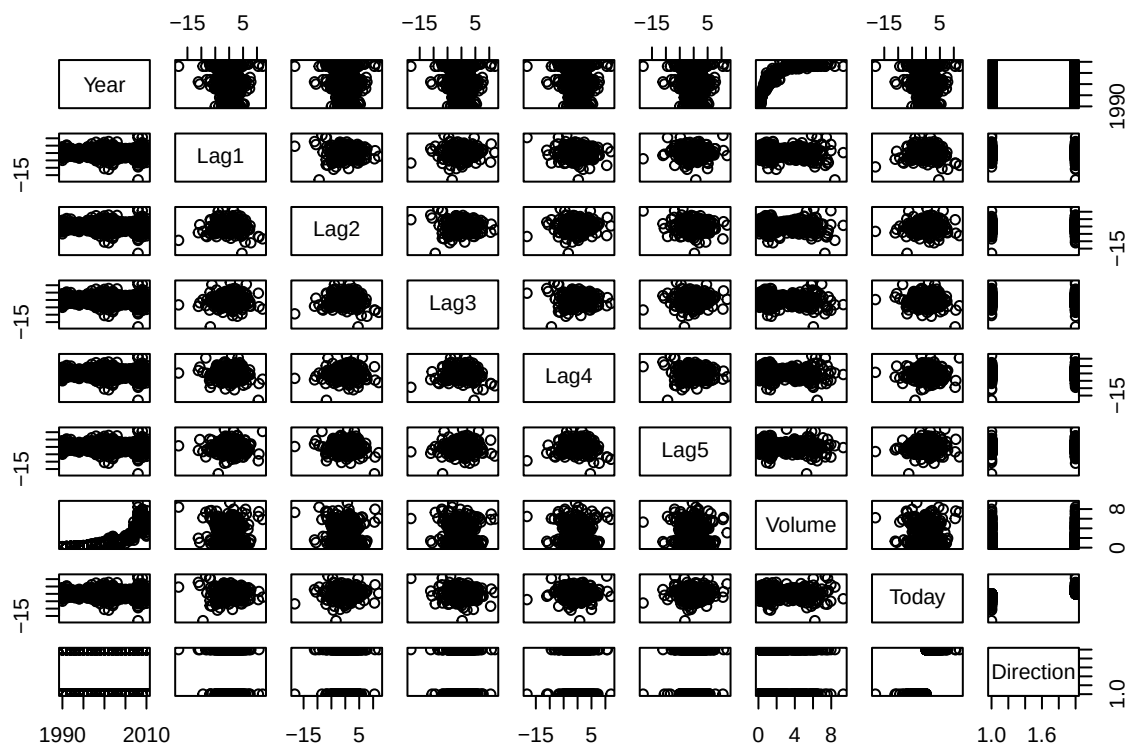
```
weekly <- ISLR2::Weekly
```

a)

```
summary(weekly)
```

```
##      Year      Lag1      Lag2      Lag3
## Min.   :1990   Min.   :-18.1950   Min.   :-18.1950   Min.   :-18.1950
## 1st Qu.:1995   1st Qu.: -1.1540   1st Qu.: -1.1540   1st Qu.: -1.1580
## Median :2000   Median :  0.2410   Median :  0.2410   Median :  0.2410
## Mean   :2000   Mean    :  0.1506   Mean    :  0.1511   Mean    :  0.1472
## 3rd Qu.:2005   3rd Qu.:  1.4050   3rd Qu.:  1.4090   3rd Qu.:  1.4090
## Max.   :2010   Max.    : 12.0260   Max.    : 12.0260   Max.    : 12.0260
##      Lag4      Lag5      Volume      Today
## Min.   :-18.1950   Min.   :-18.1950   Min.    :0.08747   Min.    :-18.1950
## 1st Qu.: -1.1580   1st Qu.: -1.1660   1st Qu.:0.33202   1st Qu.: -1.1540
## Median :  0.2380   Median :  0.2340   Median :1.00268   Median :  0.2410
## Mean    :  0.1458   Mean    :  0.1399   Mean    :1.57462   Mean    :  0.1499
## 3rd Qu.:  1.4090   3rd Qu.:  1.4050   3rd Qu.:2.05373   3rd Qu.:  1.4050
## Max.    : 12.0260   Max.    : 12.0260   Max.    :9.32821   Max.    : 12.0260
## Direction
## Down:484
## Up  :605
##
##
##
##
```

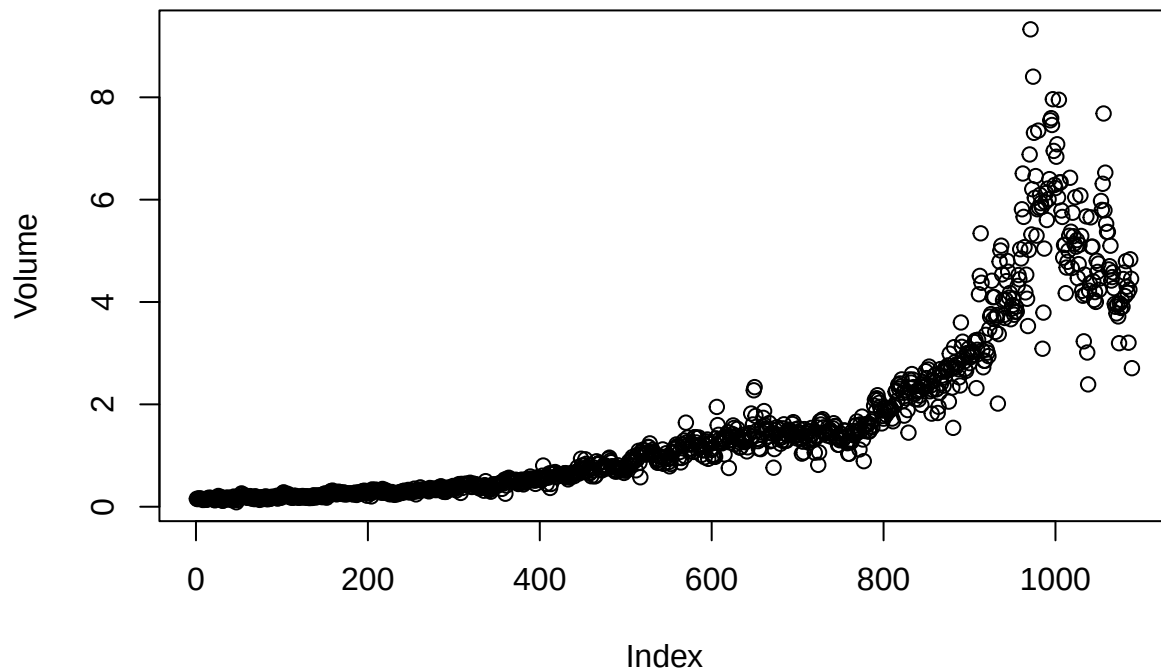
```
pairs(weekly)
```



```
cor(weekly[-9])
```

```
##           Year           Lag1           Lag2           Lag3           Lag4
## Year      1.00000000 -0.032289274 -0.03339001 -0.03000649 -0.031127923
## Lag1     -0.03228927  1.000000000 -0.07485305  0.05863568 -0.071273876
## Lag2     -0.03339001 -0.074853051  1.00000000 -0.07572091  0.058381535
## Lag3     -0.03000649  0.058635682 -0.07572091  1.00000000 -0.075395865
## Lag4     -0.03112792 -0.071273876  0.05838153 -0.07539587  1.000000000
## Lag5     -0.03051910 -0.008183096 -0.07249948  0.06065717 -0.075675027
## Volume    0.84194162 -0.064951313 -0.08551314 -0.06928771 -0.061074617
## Today    -0.03245989 -0.075031842  0.05916672 -0.07124364 -0.007825873
##           Lag5           Volume           Today
## Year    -0.030519101  0.84194162 -0.032459894
## Lag1    -0.008183096 -0.06495131 -0.075031842
## Lag2    -0.072499482 -0.08551314  0.059166717
## Lag3     0.060657175 -0.06928771 -0.071243639
## Lag4    -0.075675027 -0.06107462 -0.007825873
## Lag5     1.000000000 -0.05851741  0.011012698
## Volume  -0.058517414  1.00000000 -0.033077783
## Today    0.011012698 -0.03307778  1.000000000
```

```
attach(weekly)
plot(Volume)
```



When observing the **pairs** plot, there appears to be a logistic relationship between *Direction* and *Today*.

There appears to be little correlation between any variables, aside from a strong positive correlation between *Year* and *Volume*. In line with the **Smarket** data, there is little correlation between the lag variables and today's returns. Further, *Volume* increases over time (indexed chronologically).

b)

```
attach(weekly)
```

```
## The following objects are masked from weekly (pos = 3):
```

```
##
```

```
## Direction, Lag1, Lag2, Lag3, Lag4, Lag5, Today, Volume, Year
```

```
mod1 <- glm(
  Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 + Volume,
  family = binomial
)
summary(mod1)
```

```
##
```

```
## Call:
```

```
## glm(formula = Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 +
## Volume, family = binomial)
```

```
##
```

```
## Coefficients:
```

```
## Estimate Std. Error z value Pr(>|z|)
```

```
## (Intercept)  0.26686    0.08593    3.106    0.0019 **
## Lag1        -0.04127    0.02641   -1.563    0.1181
## Lag2         0.05844    0.02686    2.175    0.0296 *
## Lag3        -0.01606    0.02666   -0.602    0.5469
## Lag4        -0.02779    0.02646   -1.050    0.2937
## Lag5        -0.01447    0.02638   -0.549    0.5833
## Volume      -0.02274    0.03690   -0.616    0.5377
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1496.2  on 1088  degrees of freedom
## Residual deviance: 1486.4  on 1082  degrees of freedom
## AIC: 1500.4
##
## Number of Fisher Scoring iterations: 4
```

The only statistically significant variable within this model is *Lag2* at the 0.05 significance level. The *Intercept* is the only other term yielding significance within the summary of the model, at a 0.01 significance level.

c)

```
glm.probs <- predict(mod1, type = "response")
glm.pred <- rep("Down", 1089)
glm.pred[glm.probs > .5] = "Up"
```

```
# Confusion Matrix:
table(glm.pred, Direction)
```

```
##           Direction
## glm.pred Down  Up
##      Down   54  48
##      Up    430 557
```

```
# Correctly classified:
mean(glm.pred == Direction)
```

```
## [1] 0.5610652
```

Under the confusion matrix, which helps identify the class of the observations in accordance to the categorical response, *Direction*, it appears that the first model only classifies the correct *Direction* as Up or Down about 56.11% of the time. This means nearly half of the observations are incorrectly categorized, making the model slightly better than a 50/50 guessing game.

Since the model was **trained** and **tested** with the same set of observations, the **training error rate** is about 43.89%, which typically underestimates the true **test error rate**. To assess the accuracy of the model, a smaller portion of the data should be tested and the rest should be **held out** to examine a more realistic error rate once predictions are carried out.

d)

```
train <- (Year < 2009) # training data from 1990 - 2008
weekly.2009 <- weekly[!train, ] # held out data from 2009 - 2010
Direction.2009 <- Direction[!train]
```

```
mod2 <- glm(
  Direction ~ Lag2,
  data = weekly,
  family = binomial,
  subset = train
)
glm.probs <- predict(mod2, weekly.2009,
  type = "response")

glm.pred <- rep("Down", 104)
glm.pred[glm.probs > .5] <- "Up"
table(glm.pred, Direction.2009)
```

```
##           Direction.2009
## glm.pred Down Up
##      Down    9  5
##      Up     34 56
mean(glm.pred == Direction.2009)
```

```
## [1] 0.625
```

This time, using a training model from the set of observations from 1990 to 2008 and performing predictions on a held-out data set from 2009 to 2010, performs much better. This new model, using only the significant *Lag2* predictor variable, classifies the correct *Direction* around 62.5% of the time, which is surely a step up from 50/50 guessing. Further, the **training error rate** is 37.5%, which has dropped from the first model attempt of 43.89%.

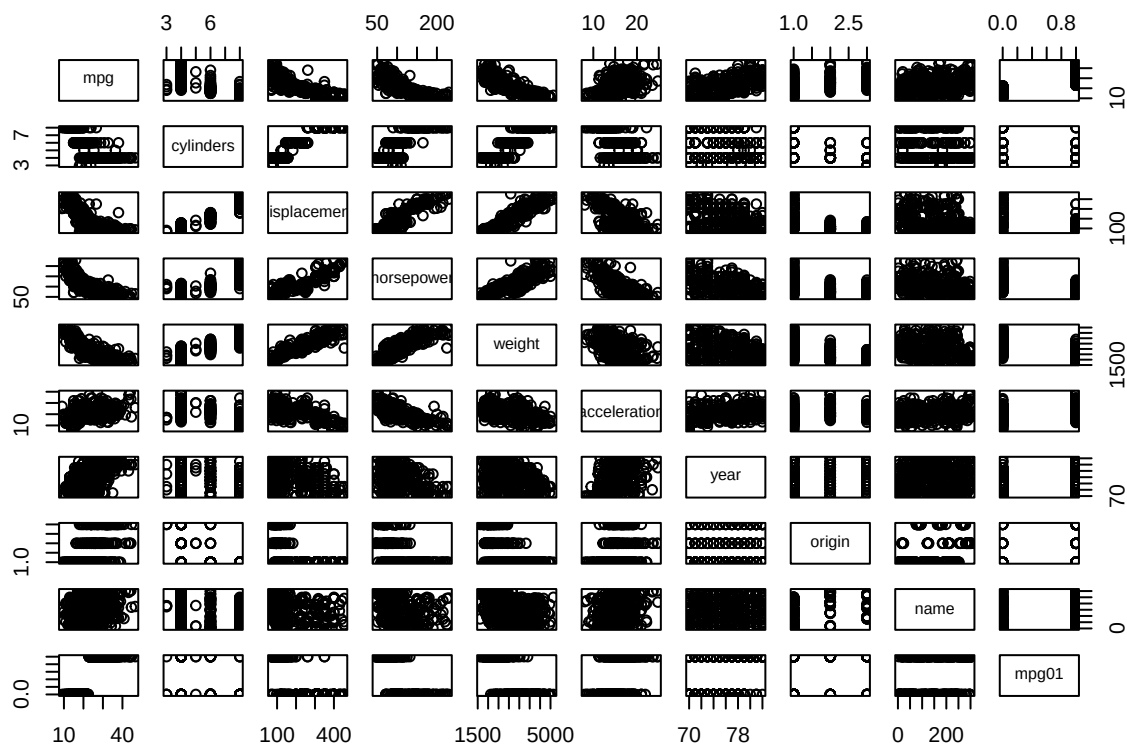
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a)

```
auto <- ISLR2::Auto
auto$mpg01 <- ifelse(auto$mpg > median(auto$mpg), 1, 0)
```

b)

```
pairs(auto)
```



Aside from *mpg*, of course, the variables that may perform well in predicting *mpg01* are *horsepower*, *weight*, and *acceleration*.

```
attach(auto)
```

```
## The following object is masked from package:lubridate:
```

```
##
```

```
##   origin
```

```
## The following object is masked from package:ggplot2:
```

```
##
```

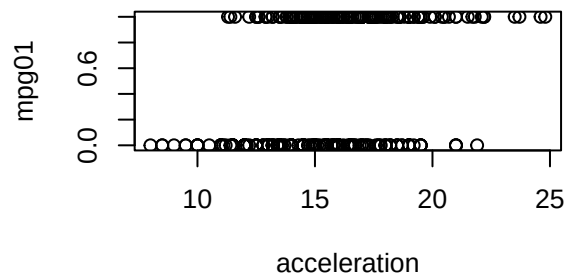
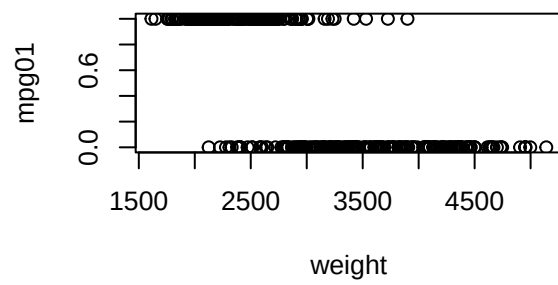
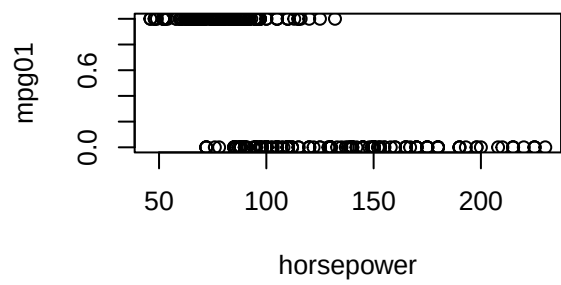
```
##   mpg
```

```
par(mfrow = c(2, 2))
```

```
plot(horsepower, mpg01)
```

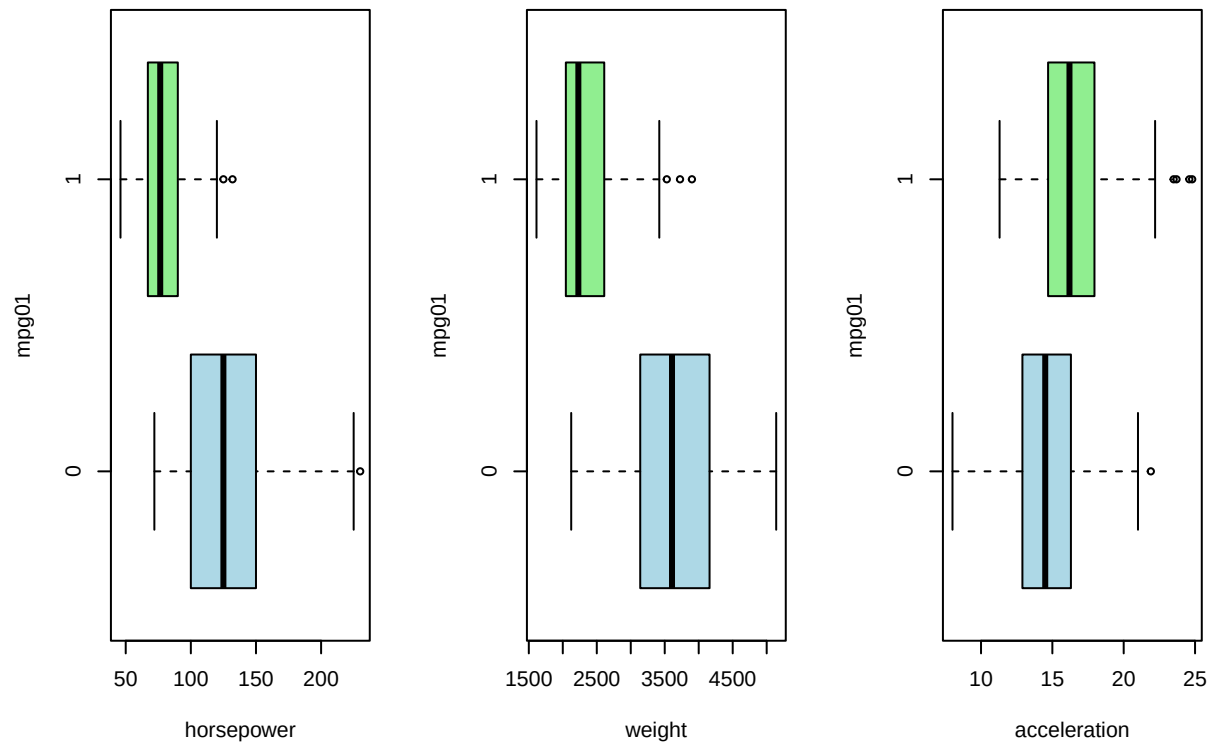
```
plot(weight, mpg01)
```

```
plot(acceleration, mpg01)
```



```
par(mfrow = c(1, 3))
boxplot(horsepower ~ mpg01,
        horizontal = T,
        col = c("lightblue", "lightgreen"))
boxplot(weight ~ mpg01,
        horizontal = T,
        col = c("lightblue", "lightgreen"))
boxplot(acceleration ~ mpg01,
        horizontal = T,
        col = c("lightblue", "lightgreen"))
```





As seen through the boxplots, it appears that the variables horsepower, weight, and acceleration each generally correspond to distinct values for *mpg01*. Note that if *mpg01* was plotted against a weakly correlated variable, such as *cylinders*, the error bars would likely overlap more, especially within the IQR.

c)

```
set.seed(5) # reproducibility

# Sample to split 80% train & 20% test
train_ind <- sample(1:nrow(auto), size = 0.8 * nrow(auto))
train <- rep(FALSE, nrow(auto))
train[train_ind] <- TRUE # training set: 313 observations

test <- auto[!train, ] # held out data (testing): 79 observations
mpg01.test <- mpg01[!train] # testing mpg01 values
```

f)

```
# Selecting variables
auto <- auto[, !names(auto) %in% c("mpg", "name")]
mod1 <- glm(
  mpg01 ~ 1,
  data = auto,
  family = binomial,
  subset = train
```

```

)
mod2 <- glm(
  mpg01 ~ .,
  data = auto,
  family = binomial,
  subset = train
)
stepAIC(mod1, direction = "both",
  scope = list(lower=mod1, upper=mod2),
  trace = F) # AIC selection

##
## Call:  glm(formula = mpg01 ~ displacement + year + weight + horsepower,
##        family = binomial, data = auto, subset = train)
##
## Coefficients:
## (Intercept)  displacement          year          weight    horsepower
## -16.158833   -0.014462     0.412873   -0.003607   -0.030852
##
## Degrees of Freedom: 312 Total (i.e. Null);  308 Residual
## Null Deviance:      432.8
## Residual Deviance: 125.7    AIC: 135.7

# Final training model
mod.train <- glm(
  mpg01 ~ displacement + weight + year + horsepower,
  data = auto,
  family = binomial,
  subset = train
)
summary(mod.train)

##
## Call:
## glm(formula = mpg01 ~ displacement + weight + year + horsepower,
##      family = binomial, data = auto, subset = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -16.158833   5.261561  -3.071 0.002133 **
## displacement -0.014462   0.007371  -1.962 0.049759 *
## weight       -0.003607   0.001032  -3.494 0.000475 ***
## year          0.412873   0.080625   5.121 3.04e-07 ***
## horsepower   -0.030852   0.017244  -1.789 0.073583 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 432.76  on 312  degrees of freedom
## Residual deviance: 125.73  on 308  degrees of freedom
## AIC: 135.73
##
## Number of Fisher Scoring iterations: 8

```

```

# Predicting test set observations for mpg01
glm.probs <- predict(mod.train, test,
  type = "response")

glm.pred <- rep(0, nrow(test))
glm.pred[glm.probs > .5] <- 1
table(glm.pred, mpg01.test) # confusion matrix

##          mpg01.test
## glm.pred  0  1
##          0 28  5
##          1  2 44

mean(glm.pred == mpg01.test) # training accuracy rate

## [1] 0.9113924

mean(glm.pred != mpg01.test) # training error rate

## [1] 0.08860759

```

The **test error** of the model predicting *mpg01* using *displacement*, *weight*, *year*, and *origin* is approximately 8.86%, which seems pretty outstanding.